Task Assignment & Roles

DEPI · Project Planning & Management · Railway Ticket Sales & Journey Performance · Oct 22 – Dec 4, 2026

6	9	9	5
Team members	Phases	Milestones	Tools

3.1 Purpose

This document defines the responsibilities of each team member in the project. It is based on the Project Timeline & Resource Plan, so every role, phase, date, and milestone matches that plan. **All team members** also define requirements, review findings, validate deliverables, prepare the final report, and present the project.

3.2 Responsibilities by team member

ME — Moustafa Elsayed

Data Cleaning

Inspect data quality, handle missing values appropriately, remove or investigate duplicates, standardize categories, and prepare the cleaned dataset.

Phase: 3. Python Data Cleaning (Oct 26 – Oct 30)

Milestone: Cleaned Dataset Ready (Oct 30)

Python · Pandas · railway.csv · railway_data_dictionary.csv

OS — Omar Saad

M Language Transformation, Database Preparation & SQL Review

Transform the cleaned data using Power Query (M Language), set up the database environment, organize tables, prepare the data for querying, and review the SQL analysis.

Phases: 4. M Language Transformation (Oct 30 – Nov 2); 5. Databricks & SQL Analysis (Nov 3 – Nov 12)

Milestones: Transformed Data with M Language (Nov 2); SQL Analysis Completed (Nov 12), with Nada Ahmed

Power Query (M) · Databricks · SQL

NA — Nada Ahmed

Data Analysis

Analyze ticket sales, ticket types, routes, journey status, delay reasons, and refund requests. SQL work is reviewed by Omar Saad.

Phase: 5. Databricks & SQL Analysis (Nov 3 – Nov 12)

Milestone: SQL Analysis Completed (Nov 12), with Omar Saad

Databricks · SQL

SA — Saja Ahmed

UI/UX Design

Design dashboard wireframes, organize the layout, and define the visual structure and user experience.

Phase: 6. UI/UX Design using Figma (Oct 29 – Nov 5)

Milestone: Figma Prototype Completed (Nov 5)

Figma

JM — Jana Mohamed

Dashboard Development

Build dashboard pages, KPIs, charts, and interactive filters.

Phase: 7. Power BI Dashboard Development (Nov 8 – Nov 20)

Milestone: Power BI Dashboard Completed (Nov 20)

Power BI

HE — Habiba Eslam

Visualization & Validation

Refine dashboard layouts, improve visual consistency, test interactions, and compare dashboard results with SQL outputs.

Phases: 7 and 8 (Nov 8 – Nov 25)

Milestone: Testing and Validation Completed (Nov 25)

Power BI · Figma · SQL results

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3.3 Task assignment by phase

#	Phase	Dates	Responsible	Milestone (date)
1	**Project Planning & Requirements**	Oct 22 – Oct 24	All team members	Project Plan Completed (Oct 24)
2	**System Analysis & Design**	Oct 25 – Nov 1	All team members	System Analysis & Design Completed (Nov 1)
3	**Python Data Cleaning**	Oct 26 – Oct 30	Moustafa Elsayed	Cleaned Dataset Ready (Oct 30)
4	**M Language Transformation**	Oct 30 – Nov 2	Omar Saad	Transformed Data with M Language (Nov 2)
5	**Databricks & SQL Analysis**	Nov 3 – Nov 12	Omar Saad (database preparation, SQL review), Nada Ahmed (analysis)	SQL Analysis Completed (Nov 12)
6	**UI/UX Design using Figma**	Oct 29 – Nov 5	Saja Ahmed	Figma Prototype Completed (Nov 5)
7	**Power BI Dashboard Development**	Nov 8 – Nov 20	Jana Mohamed (development), Habiba Eslam (visualization refinement)	Power BI Dashboard Completed (Nov 20)
8	**Integration & Testing**	Nov 21 – Nov 25	Habiba Eslam (validation), all members (review)	Testing and Validation Completed (Nov 25)
9	**Final Report & Presentation**	Nov 26 – Dec 4	All team members	Final Project Submission (Nov 29)
–	**Machine learning model (optional)**	After Nov 12	To be confirmed, only if the team decides to build it	ML Model Decision (after SQL Analysis)

3.4 Coordination

The team coordinates the outputs of each phase so that the cleaned dataset, SQL analysis results, and dashboard metrics remain consistent throughout the project. Nada Ahmed writes the SQL analysis and Omar Saad reviews it (logic, results, and consistency with the transformed data) before the SQL Analysis milestone is closed.

Data resources

- `railway.csv`: the main dataset (ticket transactions, journey details, prices, journey statuses, delay reasons, refund requests).
- `railway_data_dictionary.csv`: the reference file describing the dataset fields.

Technical resources

- Python: data inspection, cleaning, and preparation.
- Power Query (M Language): data transformation and preparation for analysis.
- Databricks and SQL: database preparation and analytical queries.
- Figma: UI/UX design and dashboard prototyping.
- Power BI: interactive visualization and dashboard development.